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# The Classification of Motor Imagery EEG Signals Based on the Time-Frequency-Spatial Feature

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**Abstract:** The effective features of the motor imagery (MI) electroencephalogram (EEG) signals plays a significant role to improve the classification accuracy for the brain-computer interface (BCI) system. Some traditional methods usually extract the frequency or spatial features without considering the related information between different channels that would affect the classification performance. This paper proposes a new method for feature extraction of EEG signals based on the fusion of time-frequency and spatial features. At the beginning, the common spatial pattern (CSP) algorithm is adopted to extract the spatial features. Then the discrete wavelet transform (DWT) and the wavelet packet decomposition (WPD) are used to extract the $\mu$ rhythm of the motor imagery EEG signals as the time-frequency features. After that, by combining the spatial and time-frequency features, the time-frequency-spatial feature is formed. Based on different kinds of features, the experimental data are classified by using the support vector machine (SVM), as well as the sparse representation classification (SRC) algorithm with the elastomeric network (EN) and L1 norm, respectively. The experimental results show that the SRC with EN has a better performance on either the time-frequency feature or spatial feature than the SRC with L1 norm does. In contrast, the SVM and the SRC with L1 norm perform better than the SRC with EN based on the time-frequency-spatial feature. The study concludes that the time-frequency-spatial feature cooperating with the certain classifiers can achieve the good classification effect for the MI EEG signals, which not only reduces the operation time but also improves the classification accuracy.

**Key Words:** Common Spatial Pattern; Feature Fusion; Motor Imagery; Sparse Representation; SVM; Wavelet Analysis



## 1 Introduction

In recent years, the BCI has been developed rapidly following the development on neuroscience, machine learning and information technologies. As a special human-machine interaction system, BCI system enables human brain to control the external electronic devices such as computers directly, rather than relying on the participation of peripheral nervous system and muscle tissue [1, 2]. The patients of paralytic cannot freely control their muscle activities that decrease their ability to communicate with the outside world. Fortunately, MI BCI technology can provide an effective way of communication for the them. The MI is the process in which the movement of specific parts of a body is repeatedly imagined in the brain [3].

Currently, the input of the BCI system is mostly the signals from the brain scalp surface recorded by a non-invasive method [4], as shown in Fig. 1. In the year of 2000, Ramoser et al. applied CSP algorithm to MI-BCI system, and found that the algorithm had a good separation effect on the features extracting from the MI EEG signals [5]. From then on, the CSP algorithm has been widely used in the research of MI-BCI system, because it can effectively extract the spatial energy features from the MI EEG signals [6, 7, 8]. Although this method can effectively extract the features of EEG signals, it is still limited to the spatial features without considering the time-frequency features of EEG signals that actually contains potential discriminative information for classification. Additionally, the wavelet analysis is a time-frequency feature extraction method, which is suitable for processing the EEG signals with non-stationary characteristics. In recent years, the wavelet analysis has been widely used in the field of signal processing. Alickovic et al. proposed a new automated seizure detection and prediction method [9]. They used discrete wavelet transform (DWT) and wavelet packet decomposition (WPD) to extract the epileptic-related features. With the effective features, we also need the good classifier for classifying. There are some emerging classification algorithms have been proposed and successfully applied to the processing of EEG signals such as SVM [10] and SRC [11]. For the MI EEG, the classification algorithm based on the sparse representation shows good separability and adaptability [12].

Fig. 1: EEG electrode cap

In this paper, we explore the classification effect based on different kinds of features by using certain classifiers. We first use the DWT and WPD algorithm to extract the $\mu$ rhythm (8-13 Hz) of the MI EEG as the time-frequency feature. Then the CSP algorithm is used for the spatial filter to extract the spatial energy features. After that, two kinds of features are fused. The time-frequency features, the spatial

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features and the time-frequency-spatial fusion features are classified by the SVM algorithm and the SRC algorithm. For SRC algorithm, L1 norm and EN are introduced as the constraints, respectively. Finally, the effect of using different kinds of features and classification algorithms are comparatively analyzed in details.

The rest of this paper is organized as follows. Section 2 provides the basic information about the experimental dataset. Section 3 illustrates our methods, including data preprocessing, feature extraction, and classification. Section 4 provides the analysis of the experimental results. The last section concludes the whole paper.

## 2 Experimental Dataset

The experimental data used in this paper is the data set III of BCI Competition II. This EEG dataset was based on the MI and derived from a 25-year-old female. The signals were recorded by three bipolar EEG channels C3, Cz and C4, as shown in Fig. 2 [13].

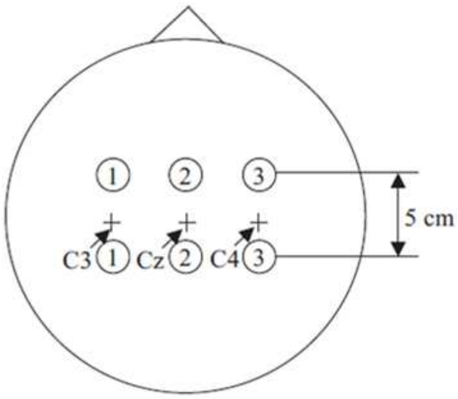



Fig. 2: Electrode positions

In this experiment, the subject imagined the left and right hand movements through the hints of an arrow on the screen, and the order of the two imaginative tasks was random. The experiment consists of 7 runs with 40 trials each. The available duration of each trail was 3-9s (as shown in Fig. 3). The EEG was sampled with 128Hz, and it was filtered between 0.5 and 30Hz. The dataset consists of 140 training samples and the remaining 140 trials as test samples, for which the left and right hand motion are labeled as class “1” and “2”, respectively.

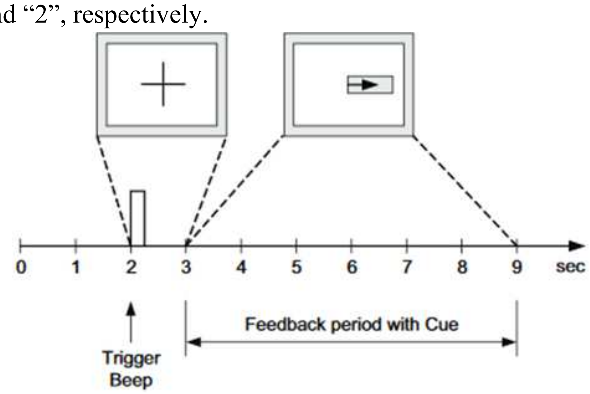


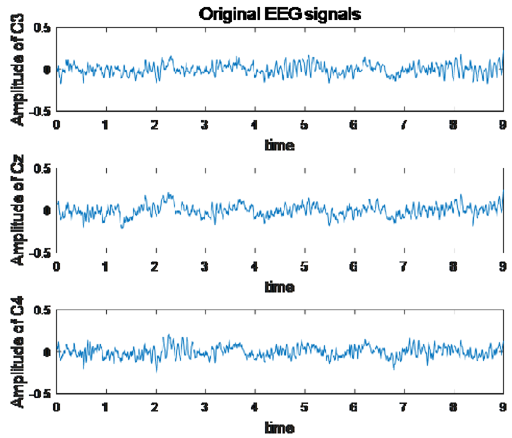

Fig. 3: Timing scheme of one trial

## 3 Methods Introduction

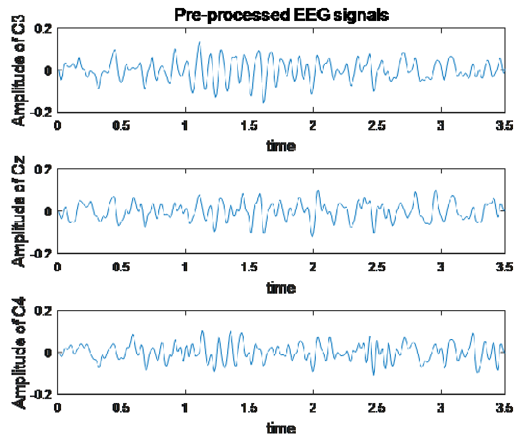
### 3.1 Preprocessing

From the experiment, we can see that the imagination time for the subject is from 3s to 9s. The average instantaneous power for left and right tasks tends to be a consistent following time. The difference of the C3, Cz and C4 channels is noticeable in 3.5s-7s [14]. Therefore, we select the EEG signals during the 3.5s-7s as the valid data for offline analysis.

In the MI-BCI system, the analysis of EEG data is mainly focused on the regions of the motor cortex, where the event-related desynchronization (ERD) and event-related synchronization (ERS) happen [15]. Therefore, extracting the ERD/ERS information in the rhythm is crucial to the MI EEG signal processing. Thus, the EEG signals are filtered between 2 and 24Hz by using a 5th order Butterworth bandpass filter.




(a) The original 0s-9s EEG data




(b) The pre-processed 3.5s-7s EEG data

Fig. 4: Comparison of original and pre-processed EEG data

In the experiment, a large amount of abnormal information can be generated in the EEG signals because of blinking, eye movement, and muscle activity. In order to reduce the impact of these exceptions, we use the Windsorizing [16] method to process the data. Fig.4(a) represents the original EEG data of the C3, C4 and Cz channels, and Fig.4(b) represents the pre-processed EEG data. It can be seen from the figures that the preprocessed EEG data are smoother and clearer than the original signals.

### 3.2 Multi-rhythm Information Extraction

#### 3.2.1 Discrete Wavelet Transform

The wavelet transform is an analysis method for signal processing in time and scale simultaneously, which not only has the characteristic of time-frequency localization, but also overcomes the shortcoming of Fourier transform. For low frequency signal, it has higher frequency resolution and lower time resolution. For high frequency signal, it has higher time resolution and lower frequency resolution. This method has good performance for the biological EEG signals that are nonlinear and non-stationary.

For any pair of real numbers $(a, b)$, a is a nonzero real number, so the formula of a continuous wavelet function is:

$$\psi(a,b) = \frac{1}{\sqrt{|a|}}\psi\left(\frac{t-a}{b}\right). \tag{1}$$

Eq. (1) is generated by the wavelet basis functions $\Psi(t)$. This function has two parameters: the scale parameter $a$ and the time parameter $b$. If the parameters $a$ and $b$ are discretized in Eq. (3.1): $a_j = 2^j; b_k = k/2^{-j}, j, k \epsilon Z$, the discrete wavelet function is rewritten as follows:

$$\psi_{j,k}(t) = 2^{-j/2}\psi(2^{-j}t - k). \tag{2}$$

In Eq. (2), $j$ and $k$ represent the frequency resolution and the time translation, respectively. So we define the discrete wavelet transform of a function $f(t)$ as:

$$C_{j,k}(f, \psi_{j,k}) = 2^{-j/k}\sum_{n=-\infty}^{\infty}\bar{\psi}(2^{-j}t - k), j, k \in Z, \tag{3}$$

where $\bar{\Psi}$ is a conjugate operation of $\Psi$. Then, the signal is restructured after the wavelet transformation, that is, the inverse process of Eq. (3) can filter and compress the signal, which is defined as:

$$f(t) = \sum_{j=-\infty}^{\infty}\sum_{k=-\infty}^{\infty}C_{j,k}\psi_{j,k}(t), j, k \in Z. \tag{4}$$

After reconstructed by Eq. (4), the signal $f(t)$ is decomposed to be some finite layers by the Mallat algorithm:

$$f(t) = A_L + \sum_{j=1}^{L}D_j. \tag{5}$$

In Eq. (5), $L$ is the number of the decomposition layers; $A_L$ is the low pass approximate component; and $D_j$ is the detail component of different scales. The whole frequency band of the signal is divided into several sub-bands. Assuming that the sampling frequency of the signal $f(t)$ is $fs$, then the sub-bands corresponding to each components are:

$$\left[0, \frac{f_s}{2^{L+1}}\right], \left[\frac{f_s}{2^{L+1}}, \frac{f_s}{2^L}\right], \dots, \left[\frac{f_s}{2^2}, \frac{f_s}{2}\right]$$

### 3.2.2 Wavelet Packet Decomposition

Wavelet packet decomposition (WPD) is an effective feature extraction method for analyzing the nonstationary signals, which provides more precise signal decomposition than that of wavelet transformation. It not only can divide the signal into multiple layers, but also decompose the high frequency band that cannot be subdivided by DWT. The basic process of WPD is below. At the beginning, the signal is projected onto the wavelet packet bases. Then we obtain a series of corresponding coefficients. Finally, the specific characteristics of the signal are described by the obtained coefficients [17]. After the signal $f(t)$ is decomposed into the $m$th layer by wavelet packet, we obtain 2m subspaces with the constant width. The sub-signals of each subspace can be represented as:

$$s_m^n(t) = \sum_{n=0}^{2^m-1}d(m,n)\psi(t), \tag{6}$$

where $d(m,n)$ represents the wavelet packet coefficients of the corresponding subspace (the $n$th frequency band in the $m$th layer). $\Psi(t)$ is the wavelet function. The energy of the wavelet packet node can be used as an effective feature since it can represent the energy of the signal. The total energy and the relative energy of the wavelet packet coefficient are defined as Eq. (7) and Eq. (8) respectively.

$$E_t = \sum_{n}^{2^m-1}|d(m,n)|^2, \tag{7}$$

$$r = \frac{E_n}{E_{total}} = \frac{d(m,n)^2}{\sum_{n=0}^{2^m-1}|d(m,n)|^2}. \tag{8}$$

### 3.2.3 Common Spatial Pattern

The CSP method maximizes the variance of the projected signals for one class, and minimizes the variance of the projected signals for another class [6]. It is suitable for two classification problems and can effectively extract spatial features from MI EEG signals.

Let $X_i$ be the MI-EEG signals with the size of $N \times S$, where $N$ represents the number of channels; $S$ represents the number of samples per channel; and $i$ represents the class of the EEG signals. The normalized covariance matrix of EEG data can be expressed as:

$$C_i = \frac{X_i X_i^T}{trace(X_i X_i^T)}. \tag{9}$$

In Eq. (9), $X_i^T$ is the transpose of $X_i$, $trace(\cdot)$ is the trace of the matrix. Here, $C_L$ and $C_R$ represent the spatial covariance matrix in the case of imagining the left hand and right hand, respectively. Thus, we can get the average value of the covariances for the two kinds of samples:

$$\bar{C}_L = \frac{1}{m}\sum_{j=1}^{m}C_L \text{ and } \bar{C}_R = \frac{1}{m}\sum_{j=1}^{m}C_R,$$

where $m$ is the number of samples per category. The mixed spatial covariance matrix for the two types of data is defined as follows:

$$C = \bar{C}_L + \bar{C}_R. \tag{10}$$

The mixed matrix $C$ in Eq. (10) is decomposed as $C = U\lambda U^T$, where $U$ is the eigenvector of matrix $C$ and $\lambda$ is the corresponding eigenvalue. The eigenvalues are arranged in descending order, and the whitening matrix $P$ is obtained according to Eq. (11):

$$P = \sqrt{\lambda^{-1}}U^T. \tag{11}$$

After whitening, the eigenvalue of $PCP^T$ is equal to 1, $C_L$ and $C_R$ are transformed as follows： $S_L = PC_LP^T$ and $S_R = PC_RP^T$, $S_L$ and $S_R$ have a common feature vector, which are expressed as:

$$S_L = B\lambda_L B^T; S_R = B\lambda_R B^T. \tag{12}$$

In Eq. (12), $B$ is a common eigenvector matrix of $S_L$ and $S_R$, and $\lambda_L + \lambda_R = I$, where $I$ is a unit matrix. Here we can find that the corresponding eigenvalue of $S_L$ is minimum when the eigenvalue of $S_R$ is maximum, and vice versa. Thus, the spatial filter is constructed by using eigenvector matrix $B$ as follows:

$$W = B^T P. \tag{13}$$

The corresponding characteristics of the MI matrix $Z_L$ and $Z_R$ are obtained by the space filter $W$ in Eq. (13):

$$Z_L = WX_L; Z_R = WX_R. \quad (14)$$

Each row of $Z_L$ and $Z_R$ is the eigenvector corresponding to $X_L$ and $X_R$, respectively. Finally, the variance of each row of $Z_L$ and $Z_R$ is used as the distinguish features to classify the two kinds of samples.

### 3.3 Classification Algorithm

#### 3.3.1 Support Vector Machine

Support vector machine is the most widely used machine-learning algorithm in classification problem. Especially for the two-classification problem, SVM has excellent classification ability and superior computational efficiency [18]. The classification formula of SVM expressed by the kernel function in low-dimensional space is:

$$f(x) = \sum_{i=1}^{n} \alpha_i y_i K(x_i, x) + b. \quad (15)$$

In Eq. (15), $\alpha_i$ is a Lagrange multiplier. $x_i$ and $y_i$ represent the eigenvectors of the training set and the corresponding class labels, respectively. $K(x_i, x)$ is a kernel function that satisfies the Mercer theorem. When different kernel functions are selected, different classifiers can be constructed. In this paper, we choose the widely used RBF kernel function:

$$K(x_i, x) = \exp(-\gamma \| x_i, x \|_2^2), \gamma > 0. \quad (16)$$

#### 3.3.2 Sparse Representation-based Classification

Sparse representation prefers to represent the data with as few atoms as possible in a given overcomplete dictionary [19]. In 2009, Wright et al. first proposed the sparse representation for classification problems [20].

We assume $y \epsilon R^m$ is the test sample and $D$ is a dictionary. If the corresponding sparse vector a can be found, the SCR model can be expressed as:

$$y = Da. \quad (17)$$

This means that the test sample $y$ can be sparsely represented on the dictionary $D$. The dictionary $D$ can be directly selected from the original training sample, that is, the predefined dictionary. We let $X_i = [x_{i,1}, x_{i,2}, \dots, x_{i,n}]$, where $X_i$ is the set of training samples from class $i$, so Eq. (17) can be transformed to $y = Xa$.

The sparse representation problem of signals can be implemented by solving a sparse regularization problem. Here, we summarize this type of problem as the following model:

$$\min_a \frac{1}{2} \| y - Xa \|_2^2 + \lambda \Omega(a) \quad (18)$$

$\Omega(a)$ is the sparse regular term. The sparse representation coefficient of the test sample $y$ on the dictionary $X$ is solved by the L1 norm minimization, as follows:

$$\min_a \frac{1}{2} \| y - Xa \|_2^2 + \lambda \| a \|_1 \quad (19)$$

The L1 norm is the sum of the absolute values of each element in a vector. L1 norm can achieve the purpose of sparse coefficients, and its operation is relatively simple. However, it is very easy to meet over-fitting problems in the actual testing process. The L2 norm can prevent over-fitting and enhance the generalization ability of the model. In this paper, we use elastomeric network constraint that combines the L1 norm and the L2 norm [8]. By this way, it can not only overcome the limitations of the L1 norm minimization constraint, but also perform continuous shrinkage and automatically select the related training samples to retain the most effective information. The model can be expressed as:

$$\min_a \frac{1}{2} \| y - Xa \|_2^2 + \lambda_1 \| a \|_1 + \lambda_2 \| a \|_2^2 \quad (20)$$

The key problem of SRC algorithm is to calculate the sparse representation coefficients. We use the accelerated proximal gradient algorithm (APG) to solve Eq. (19) and get the sparse reconstructed coefficients. APG is an effective method to solve the sparsity problem by using gradient descent. In [21], Beck and Teboulle et al. have proved that the convergence rate of APG algorithm is $O(t^{-2})$. The sparse coefficient is obtained by APG and then used to calculate the reconstruction residual $r_i(y)$ of the test sample $y$ for class $i$. We classify the sample based on the reconstruction residual: a test sample belongs to the class with the smallest reconstruction residual, as formulated Eq. (21) and Eq. (22):

$$r_i(y) = \| y - X_i \hat{a}_i \|_2^2 \quad (21)$$

$$class(y) = \arg\min r_i(y) \quad (22)$$

## 4 Data Procession and Results Analysis

### 4.1 Extraction of Features

In this study, two kinds of wavelet analysis discussed in Section 3 are used to extract the time-frequency features of the preprocessed EEG signals. When subject imagines unilateral limb movement, the amplitude of the EEG signals will change in 8-13 Hz ($\mu$ rhythm) due to the rhythmic activity in the sensorimotor cortex. Thus, we use the $\mu$ rhythm as the signal feature that is most sensitive to the right and left imagination task.

We first use the DWT to extract the feature of the EEG signals. Because the original sampling rate is 128 Hz, we decompose the EEG signals of C3, C4 and Cz channels by three-layer wavelet decomposition, respectively. This method decomposes the signals into three detail coefficients cD1, cD2, cD3 and one approximate coefficient cA1. In order to reduce the dimension of the features further, we calculate the square sum of the absolute value of the wavelet coefficients in the cD3 (8-16 Hz) to serve as the features.

When we decompose the EEG signals with wavelet packet, the extracted feature is the relative energy of the wavelet packet coefficient, which is calculated by Eq. (7) and Eq. (8). Since the sampling rate of EEG signal is 128Hz, the bandwidth of the EEG signals should be less than 64 Hz according to the Nyquist sampling theorem. In this paper, the EEG signal is decomposed by 4-layer wavelet packet, so we divide the signals into 16 bands. Because the frequency closely related to the ERD/ERS of left-right MI is the $\mu$ rhythm, the frequency band energy of node $d(4,3)$ (8-12 Hz) is chosen as the features. In this paper, db4 is used as the basis wavelet for both WPD and DWT according to Table 1.

Among many EEG feature extraction algorithms, CSP is a mature and effective spatial filtering algorithm. As

Table 1: The SVM classification results for different wavelet methods

| Method | sym3 | db4 | coif3 |
|---|---|---|---|
| WPD | 83% | 85.71% | 83.57% |
| DWT | 79% | 82.86% | 80% |

described in Section 3.2.3, for the two-classification problem, the spatial filter constructed by CSP algorithm can increase the energy of signals for one class, while decrease the energy of another class in the projection direction. By using the CSP algorithm, we also need to notice that the dimension of the features is $2m$, which varies according to the noise level of EEG data $X$ and the requirements of the classifiers. The dimension also cannot exceed the number of the selected electrode channels. Since the dataset used in this paper only selects the three channels of C3, C4 and Cz, we choose $m$ as $1$ and the dimension of the features as $2$.

Finally, we construct the time-frequency-space fusion features by combining the $280 \times 3$ time-frequency features extracted by the DWT and WPD algorithms and the $280 \times 2$ spatial features extracted by the CSP.

## 4.2 Results Analysis

This section provides the experimental analysis for the MI EEG signals. Firstly, we extract the spatial energy and time-frequency features of EEG using CSP, DWT and WPD algorithms. After that, we combine the two types of features to form the combined features as time-frequency-spatial feature. Then, for comparison the SVM algorithm and the SRC algorithm are adopted for classification. In contrast, although SRC algorithm is more complex, it has strong generalization ability and good classification effect. In the experiment, we introduce L1 norm and elastic network respectively as the constraints for SRC. However, the SRC model based on elastic network is more complicated than the SRC model based on L1 norm.

In order to further evaluate the performance of different feature extraction methods and classifiers, we adopt the performance evaluation indexes, including Precision, Recall, F1-score, Accuracy and MCC for evaluation. Table 2 to Table 4 shows the evaluation indexes of different classifiers, which are based on CSP, DWT and WPD, respectively. As a comprehensive index, F1-Score can balance the influence of Precision and Recall and evaluate a classifier comprehensively. Therefore, we focus on the comparison and analysis of the F1-score, Accuracy and MCC. As shown in Table 2, the classification accuracy of SVM, SRC+L1 and SRC+EN for the spatial features are 0.8643, 0.8786 and 0.8714, respectively. In comparison, SRC is better than SVM, and the F1-scores of SRC+EN and SRC+L1 are 0.8811 and 0.8750 respectively, indicating that the SRC+EN classification model is more robust. By comparing DWT and WPD, which are the same methods for extracting the time-frequency features, the effect of WPD method is better than that of DWT, as shown in Table 3 and Table 4. The features extracted by the two methods can achieve the highest accuracy of 0.8643 and 0.8929 with SRC+EN classifier, and the MCC values are 0.7286 and 0.7864, respectively. These results demonstrate that the WPD method is better than the DWT method when extracting the time-frequency characteristics. Above all, when only extracting time-frequency features or spatial features of MI EEG, the whole classification effect of SRC+EN is better than SVM and SRC+L1 are. In addition, we can obtain the best classification performance by using the WPD extracted time-frequency feature with SRC+EN method.

Table 2: Performance evaluation indexes for different classifiers based on spatial feature (CSP algorithm)

| Algorithm | Prec. | Recall | F1 | ACC | MCC |
|---|---|---|---|---|---|
| SVM | 0.8806 | 0.7714 | 0.8613 | 0.8643 | 0.7292 |
| SRC+EN | 0.8630 | 0.9000 | 0.8811 | 0.8786 | 0.7578 |
| SRC+L1 | 0.8514 | 0.9000 | 0.8750 | 0.8714 | 0.7441 |

Table 3: Performance evaluation indexes for different classifiers based on time-frequency Feature (DWT algorithm)

| Algorithm | Prec. | Recall | F1 | ACC | MCC |
|---|---|---|---|---|---|
| SVM | 0.8710 | 0.7714 | 0.8182 | 0.8286 | 0.6615 |
| SRC+EN | 0.8696 | 0.8571 | 0.8633 | 0.8643 | 0.7286 |
| SRC+L1 | 0.8636 | 0.7143 | 0.8382 | 0.8429 | 0.6868 |

Table 4: Performance evaluation indexes for different classifiers based on time-frequency feature (WPD algorithm)

| Algorithm | Prec. | Recall | F1 | ACC | MCC |
|---|---|---|---|---|---|
| SVM | 0.9464 | 0.7571 | 0.8413 | 0.8571 | 0.7290 |
| SRC+EN | 0.8767 | 0.9143 | 0.8951 | 0.8929 | 0.7864 |
| SRC+L1 | 0.9212 | 0.8429 | 0.8806 | 0.8857 | 0.7743 |

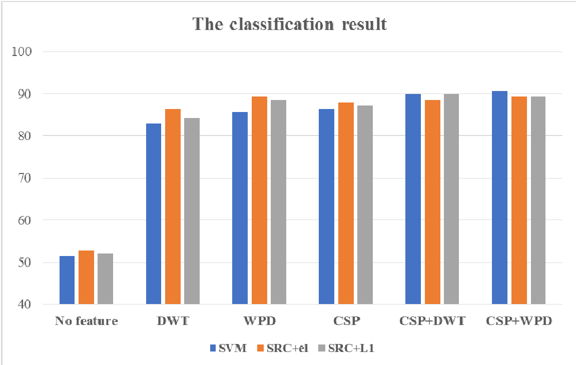



Fig. 5 Classification results for different algorithms with different EEG features

In order to verify the validity of the fusion features based on the time-frequency-spatial domain, we compared the classification results of the pre-processed EEG signals, the spatial features, the time-frequency features and the fusion features. As shown in Fig. 5, the classification of pre-processed EEG signals without features extraction cannot get a good classification result, and the classification accuracy is only about 50%. This demonstrates the importance and necessity of the feature extraction. Compared with by only extracting the spatial features or the time-frequency features, the method improves the accuracy of classification by fusing time-frequency-spatial features. The fusion features extracted by both CSP+DWT and CSP+WPD have good classification performance based on the SVM classifier, which can achieve the accuracy of 90.71%. On the contrary, the SRC+EN algorithm has the good performance for the classification of time-frequency feature or spatial feature, but it gets the accuracy of 89.29% for the fusion features, which means it is not as good as SVM and SRC+L1. Therefore, we could predict that combining the spatial feature and the time-frequency feature of the MI EEG signals can avoid our dependence on the complex classifier. Instead, we can use some relatively simple and efficient classifiers to get the better classification results.

## 5 Conclusions

The research in this paper explores the classification effect based on the MI EEG signals by extracting different kinds of features, such as time-frequency, spatial, and time-frequency-spatial features. For the time-frequency feature, we use DWT and WPD to extract the corresponding features and compare their performance for classification. The experimental results show that the time-frequency feature extracted by WPD is more effective than DWT for the classification. For the spatial feature, the CSP algorithm is adopted for the extraction. After that, the time-frequency and spatial features are combined to form the time-frequency-spatial feature (using CSP+DWT and CSP+WPD). Based on these features, the SRC and SVM classifiers are used for the classification. The classifiers SRC needs the appropriate constraint to improve its performance, so we consider the L1 minimization and the elastic network as the constraints. Compared with L1 minimization constraint, the elastic network is more flexible, but it will increase the complexity for the classifiers. Our comparative experimental results show that the SRC+EN has the better classification performance than other methods for the time-frequency feature or spatial feature. The SVM classifier obtains the better result than the SRC based on the time-frequency-spatial feature. These results show that by only using the time-frequency or spatial features, the classification methods cannot distinguish the characteristics of EEG signals enough and will affect the classification effect. By contrast, the fusion features based on the time-frequency-spatial domain could fully demonstrate the spatial and time-frequency characteristics of the MI EEG signals. With the fusion features, the classifiers not only obtain the good classification effect, but also saves the calculation time. The experimental results show the effectiveness and practicability of our proposed method, which could serve as a new way to improve the classification accuracy for the BCI system, as shown in another work [22].

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